Question			Answer	Marks	AO	Guidance
13	(a)		$H_0 : p = 0.37$ and $H_1 : p < 0.37$	B1	1.1	Allow equivalent in words Do not allow percentages
				[1]		
13	(b)		p is the probability that an adult selected at random in the United Kingdom never exercises (or plays sport)	B1	2.5	Accept 'proportion' but **not** number/amount etc B1B1 in (a)(b) if another symbol instead of p used **if correctly defined** **Underlined words needed**
				[1]		
13	(c)		$P(X) \leq 35 = 0.058\ldots$	B1	1.1	
			their 0.058.. compared with 0.05	M1	1.1	may see $P(X < 35) = 0.0386$ or $P(X \leq 36) = 0.0847$ using $X \sim B(118, 0.37)$ – SC 1 mark **Use of $P(X = 35) = 0.01961$..is M0** There is another approach using Normal approx to Binomial to find 5% CR $Y \sim N(43.66, 27.5058)$ which gives CV 35.033… Or finding value from ND $P(Y \leq 35.5) = 0.059867$.. (must be using a continuity correction so $P(Y \leq 35) = 0.0493465$.. is M0)
			0.058 > 0.05 or 35.033… > 35 or 0.059867 > 0.05 **and** 'so do not reject H_0'	A1	2.2b	'accept H_0' is ok

Question			Answer	Marks	AO	Guidance
			There is **no evidence (or insufficient evidence)** to **suggest/support** at the 5% level that the **percentage** of **adults** (selected at random in the United Kingdom) who **never exercises** (or plays sport) **is less than 37%**	**A1***	**2.4**	Fully correct contextual conclusion No assertive statements such as 'proves that' or 'shows that' 'concludes that' etc Accept percentage/proportion or probability with 0.37 Dependent on award of all other marks in (c)
				[4]		